MINIFLY RESEARCH / DESIGN PROPOSAL / 01

# Qualify byte-level parity transfer

The next phase has two independent workstreams: confirm the parity mechanism at the byte interface and measure the physical cost of retaining roughly 500 facts. The integration test follows once a reference memory configuration is selected.

## Starting evidence

The 96-run parity prototype scored 100% on both holdout suites with aligned two-step transitions. The primary one-step and shuffled controls scored 31.9% and 40.1%. The prototype took operand IDs directly, reused the distance split across worlds, omitted shortcut controls, and produced an invalid byte in the no-feedback arm. The aligned map preserves the task's alternating groups by construction.

## Input, learning, and output

Randomly assign 12 byte identities to cycle positions and independently randomize the answer-byte mapping in each world. Learn successor maps only from background byte runs, with no updates across run boundaries. Parse complete query bytes through the same causal interface in each arm. Supply the environmental teaching marker at feedback time. At test, expose only the prefix, disable writes, and score the first emitted answer plus the learner-generated newline.

## Four arms and reference checks

**A** aligned one-step plus two-step maps. **B** two one-step maps at different discounts. **C** one-step plus a randomly relabeled two-step map whose partition differs from the aligned map. **D** aligned maps with parity readout writes disabled and matched unpaired answer-byte exposure. Match actual TD update counts, feature dimensions, active units and readout size. For D, use a seeded legal-byte tie break and report answer-only chance separately from joint scoring.

Before opening sealed worlds, verify that an evaluator-only graph reference recovers the two groups and learns the answer mapping from permitted pairs. Include exact and unordered pair lookup, byte frequency, operand marginal and constant baselines.

## Holdouts and gate

Use 12 sealed worlds per suite. In the primary suite vary whole-distance holdouts by world, balance parity classes, and hold reverse orientations together. In the secondary suite withhold one identity in both operand roles while retaining it in background experience. Audit every taught and tested pair. Require at least 85% mean joint accuracy on the primary suite and at least +10 percentage points versus each of B and C, with paired 95% intervals excluding zero. Report the secondary suite separately. A pass supports narrow E3 parity transfer. Test retention through later relation learning in a separate phase.

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# Measure usable memory per byte

## Starting evidence

At 512 facts and 205 active units, widening the single store from 16,384 to 65,536 units raised old-fact retention after wave D from 50.2% to 91.6%. At 65,536 units and 410 active, post-clutter retention was 93.6% and unseen-rendering accuracy was 92.6%. The float64 readout used 128 MiB before encoder storage. The recovered encoder scans all units for top-k selection.

## Paired storage test

Use the recovered V40 SUPER arm and its V39-derived stream. Begin at 128 facts / 16,384 units / 205 active and 512 facts / 65,536 units / 410 active. Run four fresh paired worlds per cell. In each pair compare the unchanged float64 reference with float16 readout storage. Keep the encoder, stream, active count, update rule and calculation precision fixed. Compute each update in float64 and round only the touched weights on writeback. Keep float32 or int8 as separately declared follow-ups.

## Measurements and decision

Measure trained-form and held-out-rendering accuracy after D and after clutter, first-exposure transfer and revision at eight exposures. Report paired changes by world, resident readout and encoder bytes, peak RSS, and end-to-end time per byte. Count scratch buffers and any higher-precision copies. Audit that alias-2 contexts never occur in training windows. They represent the same facts and permit an E2 claim.

Proposed development gate at 512 facts: at least 90% mean old-fact retention after clutter, no more than 5 percentage points mean loss versus float64 in post-clutter old-fact retention, unseen rendering or revision, and a measured reduction in resident storage. Report paired intervals and peak memory even when the gate passes. Four worlds screen a storage format. They provide limited precision for budget decisions.

## From working points to a budget

If compact storage passes, bracket the 90% post-clutter retention boundary with loads below and above the working point at each width. Hold active count fixed for width comparisons and vary active count separately. Report a bracket wherever the sampled loads do not identify a crossing. Keep readout bytes, end-to-end memory, and time per byte visible in the budget decision. The V181 cap remains a separate constraint until a configuration is measured against it.

# Integrate after the component tests

## Mixed-stream experiment

Once a reference storage configuration is selected, pair the position-tolerant association store with a fixed all-byte predictor configuration. Use prose-wrapped fact records in an A, B, corrected-A sequence. Hold back exact query renderings before training. Probe untouched old facts, revised facts, and new facts separately, using first answer plus generated newline. Treat changed renderings of the same fact as E2.

Compare environmental teach-gated writes with a count-matched random schedule over each phase. Hold the predictor, streams, and causal read-routing rule fixed across arms. Include the predictor alone to attribute gains to the association store. Report whole-stream bits per byte, answer-site loss, recall and correction by probe family, write counts, and memory and runtime separately. Use synthetic prose wrappers first so exposures and corrections can be audited.

## Dependencies and stop decisions

**Parity and storage can proceed independently.** If the parity reference passes and the aligned learner fails, inspect whether the sparse features and readout retain the alternating partition before modifying the learning rule. If float16 misses the storage gate, preserve float64 as the reference and localize the loss before expanding the grid.

**Integration depends on selecting a memory reference.** Freeze the routing rule and quantitative integration gate after engineering checks and before final evaluation. Arithmetic stays on a separate track with an engineered positive control and relation holdouts. Defer hybrid porting while the single-store reference is being qualified.

## Execution and evidence

Freeze source hashes, seed rosters, stream generators, metric code and gates before final runs. Use separate engineering worlds to check causal parsing, read-only probes, receipts and deterministic reruns. Publish per-world records and paired summaries, with reference results and prototype results labeled by evidence level. Preserve the older V188 verdict, whose V40 reproduction gate failed overall.

## Source records

MiniFly four-seed capacity experiment and parity prototype, archive branch commit 15a78f90. Claude’s representation and generalization round-two memo and experiment notes, 2026-11-03. V188 recovered verdict. This document specifies proposed work.